

**Task 02: How far to attaining knowledge?**

Announced 02/Jun. Due: 14/Jun 10:00 PM

- Please write (only if true) the honor code. Explicitly state any source (person or thing) that you might have used, and prompts and links to LLM.
- Important: This task is solved in a team mode. If you choose to work individually, that's fine. However, the deadlines and load are considered with a team member in mind.
- Always provide explanations. Be verbose to explain your points. Your non-teammate should be able to understand you just by reading your text.
- Submission Guidelines. See [piazza](#) for standard static instructions.
- Your submission folder should look something like:

```
25v009_Task02/  
├── answers.pdf  
├── answers.tex  
├── HW.sty  
├── readme.txt  
├── ReflectionEssay.pdf  
├── images  
│   ├── bicycle.jpg  
│   ├── firstPersonView.png  
│   ├── thirdPersonView.png  
│   ├── otherImage1.png  
│   └── otherImage2.png  
├── task-body.tex  
└── 02-task-header.tex
```

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## 1 Overview

In this task, you will estimate real-world distances using a single image by – well – applying what you learnt in class. The task focuses on linearity, cross-ratio, and the practical use of object localization (using a deep learning tool) to infer distances from a single image. This topic is called single view metrology.

## 2 Background

You are provided with an image (see Fig. 1) of the way from the main building (MB) of IIT Bombay to the arch of knowledge (gyanam-paramam-dhyeyam) (“arch”). (As an aside, the arch encodes the motto of IITB).

Take a moment to digest the provided image. Observe

- the red line overlaid for reference,
- the cross road (“intersection”) that has the palm tree on one side and a building on the other side

- a bicycle

Amongst the objects, look for the street lights towards the right. We refer to these as “poles” in this assignment. The poles are numbered starting from the camera to the Arch. We will casually refer to the red line as equivalent to MB. We are going to assume that the cross road (aka intersection) is a straight line passing through the base of the third pole parallel to the red line, and that the mb-to-arch line is perpendicular to the intersection line.

### 3 Who **moved** my bicycle?

#### 3.1 Task

Consider the bicycle in Fig. 1. Report the real-world distance between the bicycle and the arch.



Figure 1: A bounding box prediction from YOLOv8.

Use a pretrained deep learning model to detect the bicycle. This is an object detection task. Please refer to tutorial here, and stick to **YOLO** models: <https://docs.ultralytics.com/tasks/detect#models> Also look at the COCO dataset <https://cocodataset.org/#home> to find out the class identifiers for bicycles and persons.

### 3.2 Questions

You are expected to write answers inline in the student space provided. (see Section 5)

1. Use a tool from Google that tells you the real world distance from MB (the red line) to the arch. Use a number in metres and approximate to the nearest distance divisible by 10.
2. Why can you NOT find the distance to the bicycle without making any assumption about the bicycle using the same tool? (Answer in one line.)
3. Ensure that you draw a line in dark blue color (using an image annotation tool of your choice) that is used for obtaining the cross reference. Show this line and the corresponding image in your submission. How did you draw the line? Keep in mind that cross-ratio is a **1D** concept and requires 4 collinear points. Make sure these points lie on the dark blue color line above.
4. Ensure that you annotate 4 points of interest as A', B', C', V' in sorted order of distance from the camera where B' corresponds to the bicycle and V' corresponds to the vanishing point. What do A' and C' correspond to? We recommend using MS Paint for annotation and identifying pixel coordinates.
5. Be sure to explain in detail your method (and code) for finding the bicycle. Does the choice of reference point within the bounding box from the deep learning model affect distance estimation accuracy? Quantify.

## 4 Your picture

In this section, take a picture of you and, alternately, your teammate (or some other person) similar to the one that has been provided in your locality. Identify two specific points of interest and tell us how far s/he is from the point of interest in the real world. We are expecting two pictures in your submission; one corresponding to the problem statement (often called as the first person view), and the second, a picture of you taking a picture of your teammate, i.e., a third person view. The second picture will show both the cameraman and the teammate (and ideally at the same time).

### 4.1 Tasks

1. Capture and include the images in your submission.
2. Detect the teammate using the same deep learning model.
3. Estimate the distance between one of the objects of interest and the team mate.

You are expected to write answers inline.

### 4.2 Question

1. As before, draw the line of interest (in blue color) marking all points of interest.
2. Explain how you got the real world distances.
3. Discuss practical challenges encountered while applying the projective geometry concepts to real-world images. Number all challenges and write independent challenges i.e. don't mix up the challenges.



4. What programs did you write and why?

## 5 Submit

Write a description of your method and all relevant aspects of your experiments. Write it in line in the text under the relevant section. Your writing should be similar to a blog post so that anyone in the class can fully understand the pain/joy/content you experienced (feel free to take copious input from an LLM but do not substitute an LLM for learning).

The total length should be no less than 500 words and should normally exceed 800 words. Your ultimate aim in writing your answers to mention important aspect of your assignment for the task, with emphasis on the last task. This will include the methods employed listing all the software tools you employed, as well experimental aspects such as when you took the picture, how many pictures you took, whether the method works in dawn, morning, and dusk, and so on. Be creative.

Do not submit any large file such as model weights and ancillary files that we can easily get from the internet to reproduce.